粒子物理　Homework　by 骆禹东 2321539.

Dirac eq:

$$i\hbar\gamma^{\mu}\partial_{\mu}\psi - mc\psi = 0 \longleftrightarrow i\hbar\gamma^{\mu'}\partial_{\mu'}\psi' - mc\psi' = 0.$$

考虑 $\psi' = S\psi$,

$$i\hbar\gamma^{\mu'}\partial_{\mu'}S\psi - mcS\psi = 0,$$

左乘 S^{-1}, 有:

$$i\hbar S^{-1}\gamma^{\mu'}\partial_{\mu'}S\psi - mc\psi = 0.$$

∴

$$i\hbar S^{-1}\gamma^{\mu'}\partial_{\mu'}S\psi = i\hbar\gamma^{\mu}\partial_{\mu}\psi$$

由于 S 中不显含变量, "S"可与"$\partial_{\mu'}$" 交换次序.

$$\gamma^{\mu}\partial_{\mu} = S^{-1}\gamma^{\mu'}S\,\partial_{\mu'}$$

其中,

$$\partial_{\mu} = \frac{\partial}{\partial x^{\mu}} = \frac{\partial x^{\mu'}}{\partial x^{\mu}}\frac{\partial}{\partial x^{\mu'}} = \frac{\partial x^{\mu'}}{\partial x^{\mu}}\partial_{\mu'}$$

∴ 可化为:

$$\gamma^{\mu} = (S^{-1}\gamma^{\mu'}S)\frac{\partial x^{\mu}}{\partial x^{\mu'}}$$

形式上, 作替换:

$$\gamma^{\nu} = (S^{-1}\gamma^{\mu'}S)\frac{\partial x^{\nu}}{\partial x^{\mu'}}$$

本题中，考虑

$$P:\ (x^0, x^1, x^2, x^3) \longrightarrow (x^0, -x^1, -x^2, -x^3).$$

$\therefore$

$$\begin{cases} \gamma^0 = (S^{-1}\gamma^0 S) \\ \gamma^1 = -(S^{-1}\gamma^1 S) \\ \gamma^2 = -(S^{-1}\gamma^2 S) \\ \gamma^3 = -(S^{-1}\gamma^3 S) \end{cases} \Leftrightarrow \begin{cases} S\gamma^0 = \gamma^0 S \\ S\gamma^1 = -\gamma^1 S \\ S\gamma^2 = -\gamma^2 S \\ S\gamma^3 = -\gamma^3 S \end{cases}$$

S 与 γ^0 对易，与 $\gamma^1, \gamma^2, \gamma^3$ 反对易

我们知道 γ_0 满足上面条件，$\therefore \gamma_0$ 为 S 的一个解.

$$\psi' = \gamma_0 \psi.$$

得证.